

EDUARDO MEDINA-ROLDAN

2025–CURRENT. GUEST EXPERT, STATE KEY LABORATORY OF ECOLOGICAL SAFETY AND SUSTAINABLE DEVELOPMENT IN ARID LANDS, URAT DESERT-GRASSLAND RESEARCH STATION, NORTHWEST INSTITUTE OF ECO-ENVIRONMENT AND RESOURCES, CHINESE ACADEMY OF SCIENCES, LANZHOU, 730000, CHINA.

EDUCATION BACKGROUND

- ❖ 1997–2002. BSc. Biological Sciences. University of Guadalajara, Mexico. Tutor: Dr. Tulio Arredondo
- ❖ 2003–2005. MSc. Environmental Sciences and Natural Resource Management. San Luis Institute of Science and Technology, Mexican Research Council (IPICYT). Tutor: Dr. Tulio Arredondo/Dr. Elisabeth Huber-Sannwald
- ❖ 2006–2012. PhD. Biology. Lancaster University, United Kingdom. Tutor: Prof. Richard Bardgett

RESEARCH AND ACADEMIC WORK EXPERIENCE

- ❖ 11/2021–05/2025. Research Fellow. National Research Council (CNR-IBE), Florence. Italy. EJP-Soil project: SERENA -Modelling Soil Functions and Soil Threats for Mapping Soil Functions and Ecosystem Services. Co-Mentor: Fabrizio Ungaro/Costanza Calzolari
- ❖ 08/2012–01/2021. Lecturer. Department of Health and Environmental Sciences, Xi'an Jiaotong-Liverpool University, Suzhou, China
- ❖ 02/2006–06/2006. IPICYT. Research Assistant. Co-Mentor: Tulio Arredondo
- ❖ 01/2003–06/2003. IPICYT. Lab and field Assistant. Co-Mentor: Tulio Arredondo

CONDUCTION AND PARTICIPATION IN RESEARCH PROJECTS

- ❖ Europe Horizon 2020. Soil Ecosystem seRvices and soil threats modElling aNd mApping (SERENA-EJP Soil). **Fellow**. 2021–2025
General Project of National Natural Science Foundation of China (42377473): Nutrient Release Effects and Driving Mechanisms of Plant Litter in Desert Grasslands in the Upper Yellow River. **Participant**. 2023–2025
- ❖ General Project of National Natural Science Foundation of China (41877540): Comparative study on the response of litter decomposition to changes in precipitation patterns in ecologically fragile arid and semi-arid areas. **Participant**. 2019–2022
- ❖ National Key Research and Development Program Project of China (2016YFC0500506): Research and demonstration of degraded grassland restoration and ecological animal husbandry technology in desert areas. **Participant**. 2016–2020
- ❖ Key Research and Development Program Project of Gansu Province(25YFWA008): Collaborative research and application of organic waste resource utilization technology in agriculture and animal husbandry between China and Mongolia. **Participant**. 2025-current
- ❖ Key Research and Development Program Project of Ningxia Province. (2020BBF02003): Research on Key Technologies and Model Creation for Enhancing the Function of Grass and Livestock Industry in Yanchi County Based on Water Resource Carrying Capacity. **Participant**. 2024–2026
- ❖ Urbanisation and heavy metal pollution in a warmer world. Xi'an Jiaotong-Liverpool University. **PI**. 2016–2017. 200,000 RMB
- ❖ Stimulated microbial activity for sustainable geotechnical remediation of unlined/uncontrolled municipal solid waste disposal sites. Xi'an Jiaotong-Liverpool University. **Co-PI**. 2013–2016. 100,000 RMB
- ❖ Soil carbon fluxes and stability of soil biological properties in response to intensified drought: plant species and restoration of degraded arid areas in Chile. Scientific and Technological National Development Fund, Chile (FONDECYT). **PI**. 2012. \$ 92,000 US

REPRESENTATIVE RESEARCH ACHIEVEMENTS

- ❖ Qu, H., **E. Medina-Roldán^{**}**, X. Wang, M. Wang, S. Wang, X. Tang. Sustainability of domestic grazing is dependent on the vegetation state of arid grasslands: Evidence from 10 years of experimental grazing intensity observations. (*Submitted*)
- ❖ **Medina-Roldán, E.[#]**, H. Huijser. Assessments to develop scientific enquiry and reflective practice in a Sino-British university (*In review*)
- ❖ **Medina-Roldán, E.[#]**, L. Kukk, R. Lorenzetti, S. Pindral, F. Ungaro, Nicolas P.A. Saby, S. Asins, A. Astover, G. Bondi, G. Buttafuoco, J. Coblinski, E. Doñate, C. Foldal, R. Hessel, D. Montagne, J. Reyes-Rojas, L. O'Sullivan, I. Cousin. Empirical estimates of net ecosystem productivity for European wheat croplands based on spatially-explicit environmental data, and their use as a climate change regulation indicator (*Submitted*)
- ❖ **Medina-Roldán, E.[#]**, G. Buttafuoco, L. Gardin, R. Lorenzetti, F. Ungaro. 2026. Comparison of national and regional assessments of soil loss rates by water erosion and soil erosion control: an application to the Tuscany region (Italy). *Land* 15: 417. [Click on doi](#)
- ❖ **Medina-Roldán, E.[#]**, M. Wang, T. Miyasaka, Y. Pan, X. Li, B. Liu, and H. Qu. 2024. Precipitation Controls Topsoil Nutrient Buildup in Arid and Semiarid Ecosystems. *Agriculture* 14:2364. [Click on doi](#)
- ❖ **Medina-Roldán, E.[#]**, R. Lorenzetti, C. Calzolari, and F. Ungaro. 2024. Disentangling soil-based ecosystem services synergies, trade-offs, multifunctionality, and bundles: A case study at regional scale (NE Italy) to support environmental planning. *Geoderma* 448:116962. [Click on doi](#)
- ❖ Qu, H., **E. Medina-Roldán^{**}**, X. Zhao, S. Wang, X. Ma, X. Tang, L. Liu., 2024. A 5 and a half years long-term rainfall experiment shows clear precipitation thresholds in litter decomposition and litter nutrient dynamics in arid and semi-arid regions. *Biol. Fert. Soils* 1-14. [Click on doi](#)
- ❖ **Medina-Roldán, E.[#]**, E. Huber-Sannwald, and J.T. Arredondo. 2013. Plant phenotypic functional composition effects on soil processes in a semiarid grassland. *Soil Biol. & Biochem.* 66:1-9. [Click on doi](#)
- ❖ **Medina-Roldán, E.[#]**, and R.D. Bardgett. 2012. Inter-specific competition, but not differences in soil microbial communities, affects uptake of different chemical forms of nitrogen by graminoids of semi-natural grassland. *PLoS ONE* 7: e51193. [Click on doi](#)
- ❖ **Medina-Roldán, E.[#]**, J. Paz-Ferreiro and R.D. Bardgett. 2012. Grazing-induced effects on soil properties modify plant competitive interactions in semi-natural mountain grasslands. *Oecologia* 170: 159-169. [Click on doi](#)
- ❖ **Medina-Roldán, E.[#]**, J. Paz-Ferreiro, and R.D. Bardgett. 2012. Grazing exclusion affects soil and plant communities, but has no impact on soil carbon storage in an upland grassland. *Agr. Ecosyst & Env.* 149:118-123. [Click on doi](#)
- ❖ **Medina-Roldán, E.[#]**, R.D. Bardgett. 2011. Plant and soil responses to defoliation: a comparative study of grass species with contrasting life history strategies. *Plant & Soil* 344:377-388. [Click on doi](#)
- ❖ **Medina-Roldán, E.[#]**, T. Arredondo, E. Huber-Sannwald, L. Chapa Vargas, V. Olalde-Portugal. 2008. Grazing effects on fungal root symbionts and carbon and nitrogen storage in a shortgrass steppe in Central Mexico. *J. Arid Env.* 72:546-556. [Click on doi](#)
- ❖ **Medina-Roldán, E.[#]**, J.T. Arredondo, E. García-Moya, F.M. Huerta-Martínez. 2007. Soil water content dynamics along a range condition gradient in a shortgrass-steppe. *Range Ecol. & Manage.* 60:79-87. [Click on doi](#)

Conference Papers

- ❖ **Medina-Roldán, E.[#]**, J. Paz-Ferreiro, J.T. Arredondo, E. Huber-Sannwald, R.D. Bardgett. Contrasting impacts of grazing on soil properties and plant communities between semiarid and temperate rangeland ecosystems. International Grassland Conference, Sydney, Australia. September 2013.

INVITED CONFERENCES

- ❖ **Medina-Roldán, E.** How to Write an academic Paper? Henan University of Engineering, School of International Education Seminar. Sept. 2023. Zhengzhou, PR of China.
- ❖ **Medina-Roldán, E.**, J. Paz-Ferreiro, J.T. Arredondo, E. Huber-Sannwald, R.D. Bardgett. Contrasting impacts of grazing on soil properties and plant communities between semiarid and temperate rangeland ecosystems. International Grassland Conference, Sydney, Australia. September 2013.

OTHER ACHIEVEMENTS

- ❖ Coblinski, J.A., S. Cornu, S. Pindral, J. Reyes-Rojas, **E. Medina-Roldán**, Lubos Borůvka, N.P.A. Saby. 2026. A multiple soil threats assessment for Europe by 2050? *Land Degradation and Development (Early view)*. [Click on doi](#)
- ❖ Qu, H., M. Wang, J. Wu; G. Wang; M. Zhang, G. Han; **E. Medina-Roldán**. Accelerating litter mass decomposition through the addition of mixed fungi in desert steppe (*In review*).
- ❖ Qu, H., M. Wang, L. Liu, S. Wang, K. Akhadi, X. Ma, S. Zare, **E. Medina-Roldán**. Litter Addition Combined with Sand Barrier is an Effective Way on Degraded Desert Steppe Restoration. *Land Degradation and Development (Early View)*. [Click on doi](#).
- ❖ Reyes-Rojas, J., D. Montagne, N.P.A. Saby, J.A. Coblinski, S. Pindral, **E. Medina-Roldán**, R. Lorenzetti, O. Scammacca, C. Piccini, L. Borůvka, and S. Cornu. 2025. Review of Soil Threats and Soil-Related Ecosystem Services European Maps: Can We Use Them to Study Their Relationships? *European J of Soil Science* 76:e70215. [Click on doi](#)
- ❖ Scammacca, O., D. Montagne, S. Asins, G. Bondi, L. Borůvka, G. Buttafuoco, A. Cadere, C. Calzolari, I. Cousin M. Czuba, C. Foldal, A. Malli, A. Klimkiewicz-Pawlas, L. Kuk, E. Lumini, **E. Medina-Roldán**, K. Michel, M.J. Molina, L. O'Sullivan, S. Pindral, E. Putku, B. Kitzler, C. Walter. 2025. Assessing and mapping soil ecosystem services and soil threats changes in agroecosystems through scenario-based approaches – a systematic review. *Sci. Tot. Env.* 966:178646. [Click on doi](#)
- ❖ Song, Z., X. Zuo, Y. Hu, H. Wang, X. Zhang, Z. Wang, J. Qiao, L. Liu, and **E. Medina-Roldán**. 2024. Plant functional traits predict soil multifunctionality under increased precipitation and nitrogen addition in a desert steppe. *Catena* 247:108486. [Click on doi](#)
- ❖ Song, Z, X. Zhao, J. Qiao, Y. Hu, X. Li, P. Yue, M. Chen, S. Wang, **E. Medina-Roldán**. 2024. Plant functional traits mediate the response magnitude of plant-litter-soil microbial C: N: P stoichiometry to nitrogen addition in a desert steppe. *Sci. Tot. Env.* 915:169915. [Click on doi](#)
- ❖ Sun, S., X. Liu, **E. Medina-Roldán**, H. Yuhui, P. Lv, Y. Xu. 2023. Precipitation regime changes alter allometry and plasticity of meristem allocation in annual herbaceous plants in a semiarid sandy grassland, Northern China. *Ecol. Res.* 38:628-638. [Click on doi](#)
- ❖ Guo, X., X. Zuo, **E. Medina-Roldán**, A. Guo, P. Yue, P., X. Zhao, J. Qiao, X. Li, M. Chen, C. Wei, and T. Yang. 2023. Effects of multi-resource addition on grassland plant productivity and biodiversity along a resource gradient. *Sci. Total Env.* 87:159367. [Click on doi](#)
- ❖ Li, X., X. Zuo, X. Zhao, S. Wang, P. Yue, C. Xu, Q. Yu, and **E. Medina-Roldán**, 2022. Extreme drought does not alter the stability of aboveground net primary productivity but decreases the stability of belowground net primary productivity in a desert steppe of northern China. *Env. Sci. & Pollution Res.* 30:4319-24328. [Click on doi](#)
- ❖ Sun, S., X. Liu, X. Zhao, **E. Medina-Roldán**, Y. He, P. Lv, and H. Hu. 2022. Annual herbaceous plants exhibit altered morphological traits in response to altered precipitation and drought patterns in semiarid Sandy grassland, Northern China. *Frontiers in Plant Sci.* 13:756950. [Click on doi](#)
- ❖ Qu, H., X. Zhao, X. Zuo, S. Wang, X. Ma, X. Tang, X. Wang, and **E. Medina-Roldán**. 2022. Litter decomposition in fragile ecosystems: A review. *Sciences in Cold & Arid Regions.* 14:151-161. [Click on journal link](#)

- ❖ Wang, S., X. Zuo, T. Awada, **E. Medina-Roldán**, K. Feng, P. Yue, J. Lian, S. Zhao, and H. Cheng. 2021. Changes of soil bacterial and fungal community structure along a natural aridity gradient in desert grassland ecosystems, Inner Mongolia. *Catena* 205:105470. [Click on doi](#)
- ❖ Zuo X, W. Mao, H. Qu, M. Chen, S. Zhao, L. Liu, S. Wang, P. Yue, X. Ma, X. Zhao, **E. Medina-Roldán**. 2021 . Scale effects on spatial heterogeneity of herbaceous vegetation in desert steppe depend on plant community type. *Ecol. Ind.* 127:107769. [Click on doi](#)
- ❖ Lv, P., S. Sun, **E. Medina-Roldán**, S. Zhao, Y. Hu, A. Guo, X. Zuo. 2021. Effects of habitat types on the dynamic changes in allocation in carbon and nitrogen storage of vegetation–soil system in sandy grasslands: How habitat types affect C and N allocation?. *Ecol. & Evol.* 11:9079-9091. [Click on doi](#)
- ❖ Lv, P., S. Sun, **E. Medina-Roldán**, S. Zhao, Y. Hu, A. Guo, X. Zuo. 2021. Soil net nitrogen transformation rates are co-determined by multiple factors during the landscape evolution in Horqin Sandy Land. 2021. *Catena*. 206:105576. [Click on doi](#)
- ❖ Zhao, Sh., T. Zhang, P. Yue, P. Lv, Y. Hu, **E. Medina-Roldán**, X. Zuo. 2021. Increased grazing intensities induce differentiation of the relationships between leaf traits and aboveground plant biomass in shrub- and grass-dominated community in desert steppe. *Ecol. Res.* 36:590-602. [Click on doi](#)
- ❖ Qu, Hao, X. Zhao, S. Wang; J. Lian, X. Tang, X. Wang, R. Zhang, and **E. Medina-Roldán**. 2021. Abiotic factors affect leaf litter mass loss more strongly than initial litter traits under sand burial conditions. *Catena* 196:104900. [Click on doi](#)
- ❖ Qu, Hao, X. Zhao, J. Lian, X. Tang, X. Wang, **E. Medina-Roldán**. 2020. Increasing precipitation interval has more impacts on litter mass loss than decreasing precipitation amount in desert steppe. *Frontiers Env. Sci.* 8:88. [Click on doi](#)
- ❖ Zhang, J., X. Zuo, X. Zhao, J. Ma, **E. Medina-Roldán**. 2020. Effects of rainfall manipulation and nitrogen addition on plant biomass allocation in a semiarid sandy grassland. *Sci. Rep.* 10:9026. [Click on doi](#)
- ❖ Hu, Y., X. Zuo, P. Yue, Sh. Zhao; X. Guo; X. Li, and **E. Medina-Roldán**. 2020. Increased precipitation shapes relationship between physiological and functional traits of *Stipa glareosa* in the grass- rather than shrub-dominated community in a desert steppe. *Plants* 9:1463. [Click on doi](#)
- ❖ Zuo, X., H. Chen, S. Zhao, Y. Ping, X. Liu, S. Wang, L. Liu, Ch. Xu, W-T. Luo, J.M.H. Knops, and **E. Medina-Roldán**. 2020. Observational and experimental evidence for the effect of altered precipitation on desert and steppe communities. *Global Ecol. & Conserv.* 21:e00864. [Click on doi](#)
- ❖ Rajasekar, A., R. Sekar, **E. Medina-Roldán**, J. Bridge, C. K. Moy, and S. Wilkinson. 2018. Biomineralisation performance of bacteria isolated from a landfill in China. *Canadian J. Microbiol.* 64:945-953. [Click on doi](#)
- ❖ Rajasekar, A., R. Sekar, **E. Medina-Roldán**, J. Bridge, C.K. Moy, and S. Wilkinson. 2018. Next-generation sequencing showing potential leachate influence on bacterial communities around a landfill in China. *Canadian J, Microbiol.* 64:537-549. [Click on doi](#)
- ❖ Paz-González, A., C.A. de Abreu, A.M. Tarquis, and **E. Medina-Roldán**. 2014. Impacts of land use changes on soil properties and processes. Editorial. *The Sci. World J.* 2014:831975. [Click on doi](#)
- ❖ Paz-Ferreiro, J., **E. Medina-Roldán**, N.J. Ostle, N.P. McNamara, and R.D. Bardgett. 2012. Grazing increases the temperature sensitivity of soil respiration. *Env. Res. Letters* 7:014027. [Click on doi](#)
- ❖ Pineda-Martínez, L.F., J.N. Carbajal, and **E. Medina-Roldán**. 2007. Regionalization and classification of climate applying principal components analysis (PCA) at northeast region, México. *Atmósfera* 20:133-145. [Click on link](#)

ACADEMIC REWARDS

- ❖ Grant for PhD studies. Council for Science and technology (CONACYT) (2006–2010)
- ❖ Grant for Master studies. CONACYT (2003–2005)

- ❖ Grant for Bachelor thesis development. IPICYT (2003–2003)
- ❖ *Grant for Bachelor thesis development.* CONACYT (2001–2002)

SELECTED CONFERENCE TALKS

- **Medina-Roldán, E.**, R. Lorenzetti (speaker), L. Kuk, S. Pindral, F. Ungaro, S. Asins, A. Astover, G. Bondi, G. Buttafuoco, J. Coblinski, E. Doñate, A. Fahy, C. Foldal, A. Matson Amanda, O. Mernagh, J. Reyes-Rojas, S. Motta, L. O’Sullivan, V. Shahrokh, E. Putku, E. Tagliaferri, E. Tonelli. An approach for mapping Net Ecosystem Productivity (NEP) as a pragmatic indicator of soil ecosystem service greenhouse gas (GHG) regulation including carbon sequestration in EU Member States. EJP-Soil Science Annual Days. June 2024. Vilnius, Lithuania (Lorenzetti speaker)
- Tarocco, P. A. Aprea, **E. Medina-Roldán (speaker)**, R. Lorenzetti, F. Ungaro. Challenges and opportunities to improve soil monitoring systems: the case of the Emilia-Romagna region in Italy. IUSS Centennial. May 2024. Florence, Italy
- **Medina-Roldán, E.**, G. Buttafuoco, L. Gardin, R. Lorenzetti, and F. Ungaro. Comparison of national and regional assessments of soil loss rate by water erosion: an application to the Tuscany region (Italy). EGU 2024. Vienna, Austria
- **Medina-Roldán, E.**, R. Lorenzetti, F. Ungaro, and C. Calzolari. Spatial relationships among soil-based ecosystem services can vary at the regional and local level. EGU. Apr. 2023. Vienna, Austria
- **Medina-Roldán, E.**, R. Lorenzetti, P. Tarocco, F. Ungaro, C. Calzolari. Complementary approaches to capture spatial relationships among soil-based ecosystem services. Soil Mapping for a Sustainable Future. Feb. 2023. Orleans, France
- **Medina-Roldán, E.** (Session Organiser). Plant soil Interactions and Sustainability. Session for Mexican Congress of Soil Science. Oct. 2022. Saltillo, Mexico
- **Medina-Roldán, E.** Effects of herbivores on the competition of plants through plant-soil feedbacks (In Spanish). Mexican Congress of Soil Science. Saltillo, Mexico, Oct. 2022
- **Medina-Roldán, E.** Herbivore-plant-soil interactions in semiarid and upland grasslands and other stories. NIOO-Knaw Seminar. Feb. 2020. Wageningen, the Netherlands
- **Medina-Roldán, E.** Herbivore-plant-soil interactions in semiarid and upland grasslands and other stories. Northwest Institute of Eco-environment and Resources, CAS. Seminar. Nov. 2019. Lanzhou, PR of China
- **Medina-Roldán, E.**, D. Hartono, and R. Sekar. Urbanisation Intensity Increases Soil Microbial Richness, But It Is Not Linked To Microbial Community Structure: A Natural Experiment In Rapidly Changing Urban Wetlands. SUITMA 10th Conference. June 2019. Seoul, South Korea
- **Medina-Roldán, E.** The patchwork assessment: a formative-summative assessment combo in an ecology module. XJTLU Annual Learning and Teaching Colloquium - Innovative Teaching and Student Learning. Apr. 2017. Suzhou, PR of China
- **Medina-Roldán, E.**, J. Grisolia, M.E. Macken, and Y. Zhang. Making the Concept of Ecosystem Services “Real”: An Interdisciplinary View from Environmental Sciences, Visual Arts and Economy. Society for Urban Ecology 2nd International Congress. July 2016. Shanghai, PR of China
- **Medina-Roldán, E.**, Wu, Y. Urban parks as models to test ecological theories. 12th ESAFS Conference, Sept. 2015. Nanjing, PR of China
- **Medina-Roldán, E.**, Jorge Paz-Ferreiro, Richard D. Bardgett. Hidden effects of large herbivores on plant competition through plant-soil feedbacks. First Global Soil Biodiversity Conference. December 2014. Dijon, France
- **Medina-Roldán, E.**, T. Arredondo, E. Huber-Sannwald. Phenotypic composition of a key grass species and soil processes in a semiarid grassland after a rain pulse event. 20th World Congress of Soil Science. June 2014. Jeju, South Korea
- **Medina-Roldán, E.** Recognizing multiple grazing impacts to use it as a management tool: the belowground perspective. SCOPE-ZHONGYU Environmental Forum. Oct. 2012. Taiyuan, PR of China

TEACHING EXPERIENCE

Modules designed/taught

Module (Yr)	Dates/Institution	Role	Highlights
Planet Earth (Y1)	09/2023, Zhejiang University of Engineering, China	Module leader (ML)	Designed & delivered for students with a basic level of English, smartphone quizzes to support learning
Ecosystem services and natural capital	03/2023, Universidad Autonoma Metropolitana, Mex.	Guest Lecturer	
Intro. Planet Earth (Y1)	2020, Xi'an Jiaotong-Liverpool University (XJTLU), China	ML	Designed & delivered. Hands-on practical collaborative assessments including one on climate system modelling with online tool
Soil and the Environment (Y2)	2013–2019, XJTLU	ML	Designed & delivered. Students self and peer assessment, co-creation of marking criteria, paper peer review, soil studio for in-class experiments, reciprocal teaching for reading primary sources; soil id field trip together with ISSAS, Nanjing
Ecology and Conservation (Y1)	2013–2020, XJTLU	ML	Designed & delivered. Patchwork text for scientific enquiry, reflective diary, role play on socio-scientific issues, peer- and self-assessments, on-line forums, video essays/video-feedback, games for ecological theory, cemetery field practice
Field Trip Skills Intro (Y1)	2013–2016, 2020, XJTLU	Co-delivered (CO)/ML	Terrestrial ecology field skills, 2020 Adjustment to e-learning with focus on collaborative home experiment for data acquisition, and plant id using online resources
Final Year Project (Y3)	2016–2017, XJTLU	ML/CO	Marking criteria for dissertations, presentations and proposals; skills development framework, supervision of FYPs
Aquatic and Urban Ecology (Y2)	2013–2017, XJTLU	CO	Designed & delivered Urban Ecology. Interdisciplinary urban ecosystem services assessment
GIS and Spatial Modelling (Y2)	2016–2017, XJTLU	CO	Designed & delivered spatial ecology. Modelling in R, practical use of GIS/RS in QGIS/GRASS. Collaborative environmental consultancy problem
Research Skills (Y2)	2013–2020, XJTLU	CO	Designed & delivered. Scientific method know-how and hands-on scientific writing
Research Skills and GIS (Y1)	2012–2013, XJTLU	CO	Designed & delivered. Hands-on practicals using GRASS/QGIS
Field Trip Skills Advanced (Y2)	2015, XJTLU	CO	Hands on practical on soil erosion/restoration, students' data collection
Environmental Chemistry (Y1)	2012, XJTLU	ML	Designed & delivered. Adjustment for students with low chemistry training
Dynamic Landscapes (Y1)	2012–2013, 2014-2015, XJTLU	CO	Co-Designed. Geo-morphological topics with in class studios
Global Environmental Issues (Yo)	2015–2018, XJTLU.	CO	Cross-university module to introduce relevant environmental topics, focus on biodiversity and evolution
Communication, Business and the Environment (Y1)	2016–2020, XJTLU	CO	Cross-university module to introduce environmental topics using a CLIL approach, focus ecosystem services and natural capital

The Anthropocene for journalism students (Y2)	2018, XJTLU	Guest Lecturer	
ATLAS (Yo) week	2016-2017, XJTLU	Coord	Adoption of sustainability activities for the intro week for freshers

Modules QC/QA as Internal Moderator (2012-2021): Assessments, structure, marking

UG Year 1	UG Year 2	UG Year 3
•Biodiversity of Animals, Plants & Microbes	•Environmental Change: Past, Present & Future	•Environmental Management
•Environmental Chemistry	•Research Design	•Environmental Impact Assessment
•Aquatic Field Skills		•Ecology in a changing world
•Dynamic Landscapes		
•Environmental Pollution		

Programmes QC/QA as Internal Moderator (2012-2021)

- ❖ New Programme in MSc. Operations and Supply Chain Management (International Business School, XJTLU)
- ❖ BSc. Environmental Sciences Program Review, Yr 2 coordinator (XJTLU-HES)
- ❖ New Programme in MDes Transdisciplinary Design (Department of Industrial Design, XJTLU)
- ❖ New Programme in BSc in Environmental Science, field-based modules designer (XJTLU-HES)

Supervision and Mentoring

- ❖ Rajasekar, A. XJTLU. PhD (co-supervisor). Stimulated microbial activity for sustainable geotechnical remediation of unlined/uncontrolled municipal solid waste disposal sites. Completed. 2018
- ❖ Supervisor UG Final Year Projects in XJTLU-HES:
 - Zhang, Y. How farmers' understanding of soil would affect the sustainable development of the landscaping industry? Completed 2019
 - Liu, Y. Impact of climate change on forest fire in Oregon State. Completed 2019
 - Xu, D. The effect of C/N ratio of Nitrogen fertilizers and precipitations patterns on the N leaching process in intact soil columns. Completed 2018
 - Hartono, D. Litter Decomposition and Soil Properties in Small Wetlands along an Urbanization Gradient. Completed 2017
 - Yue, W. The impacts of park ages on soil carbon dynamics based on the measurement of soil respiration. Completed 2015
- ❖ Academic adviser of students BSc in Environmental Sciences. XJTLU-HES. 2012–2021

TEACHING QUALIFICATIONS

- ❖ Higher Education Academy, UK fellow. 2016

- ❖ Certificate in Professional Studies in Learning and Teaching in Higher Education. University of Liverpool/XJTLU. 2016

RESEARCH STAYS

- ❖ Info&Sols Unit, INRAE, Orleans, France. Host: Nicola Saby, Iabelle Cousin, Antonio Bispo. 06/2024
- ❖ NIOO-KNAW(Netherlands Institute of Ecology), Wageningen, Netherlands. Host: Prof. Wim van der Putten. 12/2019–02/2020
- ❖ Northwest Institute of Eco-Environment and Resources, Chinese Academy of Sciences, Lanzhou, China. Host: Prof. Xiaolan Zuo. 10/2019–12/2019
- ❖ INIFAP. Professional Practices (herbarium and drylands farming/livestock outreach). Jalisco, Mexico. 04/2002–09/2002.

ADMINISTRATIVE EXPERIENCE

- ❖ SERENA project. EJP-Soil. Task coordination, whole-project admin support. 2021–2024.
- ❖ PhD Programme Director. Programme framework, liaison and students mentoring. XJTLU-HES. 2017–2021
- ❖ Research Seminars Coordinator. XJTLU-HES. 2017–2019
- ❖ Academic Practice. Quality standards in Learning & Teaching. XJTLU-HES. 2017
- ❖ Research Officer. Coordination department's research needs and support. XJTLU-HES. 2016–2017
- ❖ Exam Officer. Quality assurance on assessments based on UK system standards. XJTLU-HES. 2015–2016
- ❖ Learning & Teaching Officer. Compliance quality in learning and teaching, annual programme reviews, Learning & Teaching policy. XJTLU-HES. 2012–2015
- ❖ Student-Staff liaison Committees (Yr1, Yr2, PhD). Two-way communication channel students/staff. XJTLU-HES. 2015–2018

REFERENCES

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